



Spring Framework

Training Assignment

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RECORD OF CHANGES

No	Effective Date	Change Description	Reason	Reviewer	Approver
1	01/May/2019	Create a new assignment	Create new	DieuNT1	VinhNV
2	10/Jul/2019	Update: rename	Update	DieuNT1	VinhNV

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		TYPE	:	MEDIUM
		LOC	:	N/A
		DURATION	:	120 minutes

Unit 1: Spring Framework Core

Objectives:

After finishing the following exercises, trainees will:

1. Understand and practice with OOP, IO
2. Understand and practice with Spring core (IoC, DI)

Problem Descriptions:

We are beginning to build our Learning Management System (LMS). This task involves creating the core Course entity and a service to save/load course data from a file, all managed by the Spring container.

Task 1: Create the Course Entity

Write a class named **Course.java** in the model package that implements Serializable and has:

1. Four attributes: String title, String instructorName, String description, int durationHours.
2. One constructor with all attributes as parameters.
3. One function: void **displayInfo()**. For example: The course "Spring Framework" will print: [Course: Spring Framework] by John Doe (40 hours): Learn Spring Core, MVC, and Boot.
4. Override the **toString()** method as follows:

```
@Override
public String toString() {
    return "Course [title=" + title + ", instructor=" + instructorName + ", duration=" + durationHours + "];"
}
```

Task 2: Create the CourseService

Write a class named CourseService.java in the service package that has:

a. Function void **saveCourses**(List<Course> courses): This will save the list of courses passed in the parameter to a file named courses.dat.

(Hint: Use ObjectOutputStream).

b. Function List<Course> **getCourses**(): This will read the courses.dat file and return its content as a List<Course>.

(Hint: Use ObjectInputStream).

Task 3: Create the Test Class

Write a class named Test.java in the main package that has:

- a. Attribute: CourseService courseService;
- b. A main method, where you should:
 - Create a list of Course objects (e.g., 2-3 courses).
 - Call the saveCourses() method from courseService to save this list to the file.

- Call the getCourses() method from courseService to get the list of saved courses.
- Loop through the retrieved list and call the displayInfo() method for each course.

Note: You must configure Spring to create a bean for the CourseService class and inject that bean into the courseService attribute of the Test.java class (using either XML configuration or Annotation-based configuration).

Technical Requirements:

- JDK 1.8 +
- Maven 3 +
- Spring Framework Core

-- THE END --